



Dust in the errors: Comparative econometric measurements on Mexico's Hoy no Circula

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April 2025

Abstract

Regression Discontinuity Designs (RDD) often have measurement error in their running variables, which can lead to invalid inferences if not properly addressed. In this study, we replicate and extend Davis's (2008) analysis of car restriction policies and air quality in Mexico City by integrating Dong and Kolesar (2023) methodological recommendations —bias-aware confidence intervals and donut trimming - alongside our own choice of optimal bandwidth selection to address measurement error. Additionally, we employ semi-parametric estimation as an external validation of our approach. Under very conservative bias adjustments, our results remain inconclusive; however, when more realistic assumptions are imposed, we document a statistically significant reduction in air pollutants in the short run (weeks after the policy was implemented), with subsequent evidence of a rebound effect afterward. Our enhanced framework not only refines causal inference in contexts with imprecisely measured running variables but also offers both theoretical and empirical advancements for RDD applications in similar scenarios.

Introduction

The Regression Discontinuity (RD) design has been widely used since the late 1990s and recognized for its internal validity to estimate local average treatment effects (Davezies 2017). RDDs normally rely that assignment to treatment or control is through a continuous running variable being on either side of a fixed threshold. However, an econometric challenge is that the running variable that distinguishes between control and treatment groups often has measurement error. By some estimates, 23% of over 150 recent Economic papers used a running variable measured with error in leading journals (Dong & Kolesár, 2023). This poses a validity problem to RD design experiments. Measurement error, if not solved, leads to inconsistent estimates for the average treatment effect at the threshold.

There exist multiple proposed solutions to the common question of measurement error in the running variable. Several methods rely on auxiliary assumptions about the measurement error form or additional population data (Dong & Kolesár, 2023). As a result, some methods are often not possible with available information or based on flawed assumptions. Therefore, Dong & Kolesár (2023) propose a new method to still be able to identify a causal relationship under the usual RDD assumptions despite the measurement error. Assumption 1 is that the measurement error does not affect the treatment allocation. Assumption 2 is measurement error is not correlated with the outcome variable. If both assumptions hold, identifying a causal relationship is possible. Nevertheless, this causal relationship will be valid for the grouped or rounded unit of measurement of the running variable for which we observe data and not for the ideal unit of measurement without error. The authors suggest that ignoring the measurement error nonetheless yields an estimate with a causal interpretation: the average treatment effect for units whose observed running variable equals the cutoff (Dong & Kolesár, 2023).

“The effect of driving restrictions on Air quality in Mexico city” (Davis, 2008) was chosen as our empirical application and was also mentioned in Dong and Kolesar’s appendix. Davis’ paper investigates whether restricting potential car usage by 20% every day reduces air pollution. According to his paper, the policy was ineffective. We suspect this might not be a precise answer as air pollution can remain in Mexico City for at least one week (De Foy et al., 2006). This persistence may violate the assumption of correct treatment assignment timing, potentially leading to non-causal estimates. Furthermore, other papers replicating Davis dataset have given different answers using different methodologies than this paper (Salas, 2010; Gallego et al. 2013). According to Dong & Kolesár (2023), one might still identify a causal relationship for the average air pollution per week if the assumptions (1) and (2) are satisfied.

If assumption one is partially violated, they propose to use a doughnut design-approach. Observations for which the measurement error has affected the treatment allocation have to be removed. The RDD will be estimated based on the dataset without the removed observations. Furthermore, they also suggest using bias-aware confidence intervals. Bias-aware confidence intervals take the bias produced by the possibly wrong estimation of the continuous function

before and after the cutoff into account.

An advantage of bias-aware inference is that the estimator and CI construction remains the same whether the support of the running variable is continuous, discrete, or otherwise irregular, and whether the running variable is point or partially identified (Dong, 2023). The disadvantage is that Dong and Kolesar's method is best relevant when the support of the measurement error is relatively narrow (which includes most settings with rounding or grouping error); otherwise, the doughnut trimming may end up removing too many observations and preclude informative inference (Dong, 2023). There exists a trade-off between inference and efficiency in the method. We apply this extension to Davis's dataset where we found promising results.

The impact of HNC on air pollution will be estimated in our extensions where we first apply bias-aware confidence intervals with donut trimming to Davis Dataset and then use Imbens & Kalyanaraman's (2012) approach to calculate the optimal bandwidth. Whereas Davis found little to no impact of HNC on air pollution, our econometric approach shows that there was an impact in the immediate months where air pollution (as measured by common transport emissions reduced) before rebounding in the years after that we believe is caused by people reacting by buying extra and dirtier cars. Our approach as detailed in this paper was more sensitive to measurement error on two levels. The first is that we take into account air pollution is an imprecise measure of car emissions as supported by relevant literature so relied on donut trimming and bias-aware confidence intervals. The second level was the optimal bandwidth that was more reflective of time as a variable in emissions over several months and differentiated between an initial drop in emission and a rebound months after that the original paper of Davis had not identified. Combining the two approaches, our results were both empirically and theoretically superior as the more conservative uncertainty approach (having bias-aware confidence intervals and donut trimming) was counteracted by the greater sensitivity of our optimal bandwidth.

Methodology

The theoretical article that is central in this paper is the academic article of Dong & Kolesár (2023) called "When can we ignore measurement error in the running variable?". This theoretical paper builds on the existing literature on measurement error in the running variable and investigates some conditions under which such errors can be corrected for. It further examines to what extent the estimated coefficients can still be given a causal interpretation after applying these corrections. Measurement error in the running variable is a considerable problem in regression discontinuity research, as it smoothens out any discontinuity at the cutoff (Davezies & Le Barbanchon, 2017). One often corrects for this by using auxiliary datasets to recover the true running variable without measurement error. Nonetheless, in many or even most cases such auxiliary datasets are not available. Dong & Kolesár (2023) show in their paper that one can still recover a causal estimate under some conditions and despite the measurement error if

one uses a doughnut design and bias aware confidence intervals.

More concretely, they argue that there are two main high level conditions for which it is still possible to recover a causal interpretation despite a measurement error. First, there should be a correct classification of the treatment assignment. In other words, the measurement error should not be correlated with the treatment assignment. Second, there should be a continuity of the conditional expectation function. The expected potential outcomes have to be continuous at the cutoff. The measurement error cannot be correlated with the potential outcomes.

Furthermore, Dong & Kolesár (2023) also identify low level conditions which are sufficient conditions that ensure the high level conditions to be true. First, doughnut trimming around the cutoff might eliminate any correlation between the measurement error and the treatment assignment. However, it is important to not remove too many observations in order to not bias the results. Second, the measurement error distribution has to be smooth. These high level and low level conditions are typically satisfied by grouping errors (like for instance due to rounding), classical measurement errors (measurement error uncorrelated with the potential outcomes and true running variable), Berkson measurement errors (measurement error uncorrelated with the potential outcomes and the observed running variable), and heaping.

If the high level conditions and the low level conditions are satisfied, causal identification can be recovered for a continuous, discrete, and mixed running variable. However, one should use bias aware confidence intervals to take the bias produced by the possibly wrong estimation of the continuous function before and after the cutoff into account. This is especially true for discrete or mixed running variables.

Moreover, Dong & Kolesár (2023) also argue that their correction method also applies to fuzzy regression discontinuity designs (RDDs where the uptake is less than 100%). Nevertheless, in the case of a fuzzy design additional conditions are needed. First, there has to be monotonicity in treatment assignment. In other words, there cannot be any defiers. Second, there has to be continuity in expected outcomes and in the first stage of the treatment probability. If no treatment had taken place, the expected outcomes and treatment probability should be continuous at the cutoff. Third, the measurement error should be independent from the latent treatment process. To put it differently, the measurement error should not systematically affect whether an individual is a complier, never-taker or always-taker.

Under the above mentioned conditions, it is possible to recover a causal interpretation of a RDD with measurement error on the running variable. However, it is important to note here that this causal interpretation applies to the observed running variable (that is for instance rounded) and not for the ideal running variable (for instance the unrounded running variable).

Dong & Kolesár (2023) illustrated their new correction method by also providing an empirical section in their article. More specifically, Dong & Kolesár (2023) replicated the research of Holbein and Hillygus (2016) who studied the impact of preregistration on youth voter turnout in Florida. Young people who were too young to vote in the 2008 election were given the opportunity to preregister for future elections. The running variable was the date of birth and

the cutoff November 4, 1990. It was expected that people just after the cutoff would have been more likely to vote in the next elections because of their preregistration. Nonetheless, the running variable contained a measurement error. The date of birth was recorded in months and not in days (rounding error). Hence, Dong & Kolesár (2023) made use of doughnut trimming in order to satisfy the higher level conditions. Furthermore, they computed bias-aware confidence intervals. Therefore, they were able to recover the causal estimate of the preregistration. However, it is important to note that this estimate applies to birth dates recorded in months rather than in exact days.

Replication Paper Summary

In the appendix B of the academic article of Dong & Kolesár (2023), Dong & Kolesár (2023) identified five research papers for which the measurement error on the running variable was not yet corrected for and that could almost surely be correct for with their suggestions. We contacted the authors of four of the papers and requested for the data to replicate findings. Davis (2008) and Stancanelli & Soest (2012) replied and provided their research data. Eventually, we decided to replicate the research of Davis (2008), since the measurement error was clearer in his research.

The research of Davis (2008) is about the implementation of “Hoy No Circula” in Mexico City. In the 1980’-90’ Mexico City used to be called the most polluted city on earth with almost all the air quality measures tremendously exceeding the WHO guidelines. Hence, the Mexican government tried to tackle this problem and implemented “Hoy No Circula” on November 20, 1989. “Hoy No Circula” limited randomly the usage of some cars by stating that license cars beginning with a certain number were not allowed on one weekday each week between 5h and 22h. The reasoning was that this policy would be easy to enforce (checking license plates is not complicated) and affect people randomly while effectively reducing car emissions. By making use of data from the Mexican Automated Environmental Monitoring Network Davis (2008) analyzed the effectiveness of the policy with an RDD design. More specifically, he estimated this equation:

$$y_t = \alpha_0 + \beta_1 HNC_t + \alpha_2 x_t + u_t$$

The outcome variable is the weekly average pollution. The estimated parameter β_1 of HNC indicates the impact of the policy. HNC is a dummy equaling one if the observation was after November 20, 1989. “The vector of covariates, x_t , includes month of the year dummies and quartics in temperature, relative humidity, and wind speed. These variables are included to control for the seasonal pattern of air pollution. The air quality equation is estimated using least squares (OLS) for four different time windows ranging from 1986-1993 to 1989-1990” (Davis, 2008; p. 8). Furthermore, Davis (2008) also estimated the equation with high-order

polynomials and kernel local averaging.

According to their estimated parameters, the policy was not effective. No significant discontinuity was observed and Davis (2008) provided explanations why this might be the case (increased second hand car consumption and increased usage of heavily polluting private public transport).

Literature review around Davis

One of the first scholars who studied “Hoy No Circula” (HNC) were Eskeland & Feyzioglu (1997). Based on a log linear regression model they studied the impact of HNC on the demand for fuel. Surprisingly, they found that the introduction of HNC increased the demand. Furthermore, they also had a look at car registrations and car sales data. Before the introduction of HNC Mexico city used to export cars to other parts of the country. After the introduction, however, they started to net-import cars. Both the sales and the registrations increased significantly.

After the publication of Davis’ (2008) article, Salas (2010) tried to replicate the exact same results. Interestingly, they found that a slight change in methodology had a considerable effect on the sign of the estimates and the significance of the estimates. Davis (2008) made use of a seventh, eighth, and ninth order polynomial in order to estimate the RDD. Nevertheless, they show that if one uses instead a fifth order polynomial three of the five pollutants turn out to have reduced significantly in the RDD design. Moreover, Davis (2008) only used a time window of eight years for the RDD analysis. Nevertheless, Salas (2010) showed that the results become more negative and more significant when one makes the time window smaller. In a six year time window, two pollutants increased significantly, whereas in a two year time window, two pollutants decreased significantly. Nonetheless, Salas (2010) confirm the results of Davis (2008) in the long run. In the long run they found as well that the policy was not effective, most likely due to an increase in car consumption and a change in driving behaviour.

In the shorter run, Salas (2010) chose to estimate the RDD by making use of a Kernel regression and Lowess smoothening (which is a RD with a shorter time frame). In contrast to Davis (2008), they found a significant reduction in carbon emissions ranging between 12% and 18% in the first months.

Subsequently, Gallego et al. (2013) analyzed a similar air pollution policy in Santiago (Chile) and reanalyzed the HNC in Mexico City. However, in contrast to Davis (2008) they made use of the Imbens and Kalyanaraman method which is based on local linear regressions near the policy implementation threshold (Imbens & Kalyanaraman, 2012). Moreover, Gallego et al. (2013) also had a look at car registrations, the impact of the policy per income category, gasoline sales, traffic flows, and taxi license prices. They found that there was a significant increase in car ownership eight to twelve and a half months after the HNC introduction. Moreover, they found that middle income groups were the most affected by the policy change. Richer households were able to afford multiple cars. Furthermore, Gallego et al. (2013) found

a significant long run increase in pollution of around 20% which became visible after one year of the implementation of HNC. Nonetheless, they also found a significant short run reduction. Hence, they consider this kind of pollution measure only effective if it is implemented in a very short period of time and in emergency cases.

Finally, Blackman et al. (2015) tried to quantify the average costs of “Hoy No Circula” (HNC) for individual households. By making use of the contingent valuation method, they found that the average cost was around 77 dollars and 103 dollars per car per year which represented one percent of the median income of a household. The societal costs were estimated to be between around 367 million dollars and 490 million dollars per year. Blackman et al. (2015) found these results by randomly asking drivers whether they would pay 50 dollars, 70 dollars etc. for a HNC exemption of one year. Subsequently, they used the results of this experimental survey to calculate the average willingness to pay based on probit models.

Data Description and Application

In this section, we summarize the data sources and variables for our replication of Davis’s (2008) analysis of Mexico City’s Hoy No Circula (HNC) program. We rely on two primary sources. First, we use the dataset and code shared by Professor Lucas W. Davis on his personal website, which clarify how he compiled and processed daily pollution levels, weather conditions, and policy details.

Second, the original source of Mexican air qu Mexico City’s Atmospheric Monitoring System (SIMAT), which provides hourly readings of key pollutants—carbon monoxide (CO), nitrogen dioxide (NO), ozone (O), nitrogen oxides (NO_x), and sulfur dioxide (SO₂)—from 1986 to the present. These readings can be averaged across stations and over time to produce daily or citywide pollution measures.

Davis’s original study focuses on the period from 1986 to 1993, capturing pre- and post-HNC trends after its introduction in November 1989. Our dataset also focuses on the time period with the intuition of understanding the short run (measured in weeks) and Long run (measured in years) impact of the policy. We then had multiple objectives in replicating and extending Davis’ original research. Rather than simply copying Davis and other researchers code, we instead focused on their econometric approaches, and used a composite approach to what we consider is more sensitive, more accurate and more careful regarding potential biases as to the impact of HNC on air quality in Mexico.

1. Replicate Davis results with required controls
2. Account for Measurement error (donut trimming and bias-aware CI)
3. Improve sensitivity and accuracy of impact (Optimal bandwidth)
4. Validate with semi-parametric alternative method

1 Replication of Davis in Stata and R

Since Davis (2008) provided the Stata code and data on his website, it was fairly simple to download his package. Nonetheless, it was very challenging to convert the Stata file into a working R file, because Davis' (2008) code is very densely written and with commands from the Stata version of more than fifteen years ago. The first thing we tried was to replicate the RDD analysis with the ninth-order polynomial in Stata. More specifically, we estimated the following regression for a time window of eight years:

$$\begin{aligned} \log(x_t) = & \beta_0 + \beta_1 \text{hnc}_t + \sum_{i=1}^9 \beta_{1+i} t_{\text{scaled},t}^i \\ & + \sum_{j=1}^4 (\gamma_j Wtmp_t^j + \gamma_{j+4} Wtmp_{t-1}^j + \delta_j Wwsp_t^j + \delta_{j+4} Wwsp_{t-1}^j + \theta_j Wrh_t^j + \theta_{j+4} Wrh_{t-1}^j) \\ & + \phi_{\text{month}} + \phi_{\text{dow}} + \phi_{\text{hour}} + \phi_{\text{hourweekday}} + \varepsilon_t \end{aligned} \quad (1)$$

As already mentioned, `hnc` represents the treatment (equals one after the introduction of the program), `t` represents the time elapsed before/after the treatment (up to a ninth polynomial has been included). Furthermore, we included weather controls and their one-period lags, each modeled up to a fourth-order polynomial, represented by `Wtmp`, `Wwsp`, and `Wrh`. Finally, fixed effects for the months, day of the week, hour, and hour weekdays interactions were included. The coefficient of interest is β_1 which represents the effect of the policy.

As shown in the table below, we obtained results that differ from those reported by Davis (2008), despite using the exact same code and dataset provided on his website. However, our estimated coefficients and standard errors align perfectly with the findings of Salas (2010), who also replicated Davis (2008). It is important to note that the estimated coefficients from both Davis (2008) and our replication fall within the respective standard errors.

Outcome	Coefficient	Standard Error
logco	0.0079	0.0919
logno2	-0.0578	0.1641
logo3	0.0531	0.1038
lognox	0.0006	0.1342
logso2	0.2073	0.1218

Table 1: Replication results of Davis (2008) in Stata

Subsequently, we tried to do exactly the same in R instead of Stata. For unclear reasons, the standard errors of the same regression in R with the package `fixest` and `lfe` were very

high. However, with the package `glm` the regression generated almost the same coefficients and standard errors as in Stata (see the table below).

Outcome	Coefficient	Standard Error
logco	0.010	0.090
logno2	-0.060	0.140
logo3	0.050	0.100
lognox	0.000	0.110
logso2	0.210	0.130

Table 2: Replication results of Davis (2008) in R

To conclude, our replication of Davis (2008) aligns with Salas (2010) both in Stata and in R. Since the authors of this paper were more proficient in R than in Stata, the following extensions on the replication of Davis (2008) were all conducted in R. Furthermore, most of the extensions were done by making use of the package `glm`.

2 Extension 1 (Doughnut + Bias-Aware CI)

As suggested by Dong & Kolesár (2023), the first extension of the replication of Davis (2008) consists of doughnut trimming and bias-aware confidence intervals. As already mentioned, the doughnut trimming is conducted to ensure correct classification of the treatment assignment. Since air pollution can stay for one week in the valley of Mexico City, the first week violates the correct treatment assignment assumption. Hence, in this first extension we will remove those observations. Furthermore, we will compute the bias-aware confidence intervals which also consider the bias generated by the choice of the functional form.

The bias-aware confidence intervals cannot be solely estimated by the current dataset and rely on some assumptions. As explained in Kolesar (2024) in 'Honest inference in RDD', bias-aware inference methods bound the potential bias of the estimator by making an explicit assumption on the smoothness of f_y . One method is to restrict the curvature of f_y by its second derivative by a chosen parameter M . The values of $M = 0.1$ and $M = 1$ were chosen where $M = 1$ is a very conservative estimate with a far larger confidence interval and $M = 0.1$ is more likely with the evidence given by researchers who previously worked on the Davis Dataset. In real terms, $M = 1$ roughly means there was a 50 % jump in air quality measurement data in the discontinuity which seems extremely unlikely and cannot be found in any data. An estimate of $M = 0.1$ is a much smaller decrease of roughly 5 % in Mexico city at the discontinuity which seems more realistic especially as only a realistically small amount of vehicles would be actually affected under the HNC policy. While we kept $M = 1$ as a conservative estimate, $M = 0.1$ estimates of the confidence intervals are more realistic.

The table below shows the results for the trimming of one week, two weeks, and three weeks with robust standard errors clustered in blocks of 35 days, bias-aware confidence intervals for $M = 1$ and $M = 0.1$. One observes clearly that the conservative bias-aware confidence intervals of $M = 1$ are much wider than the confidence intervals of $M = 0.1$ and the robust clustered standard errors. The bias-aware confidence intervals of $M = 0.1$ are considerably smaller than the two other confidence intervals. The more weeks that are trimmed, the higher the point estimates become. However, this increase in point estimates is only significant for NOX when trimming for two weeks and three weeks for the confidence intervals of $M = 0.1$. The point estimates for SO2 are significantly positive regardless of the doughnut trimming for the confidence intervals of $M = 0.1$.

In the following extensions we will use the doughnut trimming of one week based on the research of De Foy et al. (2006) on air pollution in Mexico City.

Outcome	Davis (GLM)	One Week	Two Weeks	Three Weeks
logco	0.010	0.010	0.040	0.040
	(-0.166, 0.186)	(-0.166, 0.186)	(-0.156, 0.236)	(-0.156, 0.236)
	(-0.510, 0.520)	(-0.510, 0.530)	(-0.480, 0.550)	(-0.470, 0.560)
	(-0.060, 0.070)	(-0.060, 0.080)	(-0.030, 0.100)	(-0.020, 0.110)
logno2	-0.060	-0.040	-0.010	0.040
	(-0.334, 0.214)	(-0.314, 0.234)	(-0.304, 0.284)	(-0.234, 0.314)
	(-0.570, 0.460)	(-0.560, 0.480)	(-0.520, 0.510)	(-0.480, 0.560)
	(-0.120, 0.010)	(-0.110, 0.030)	(-0.070, 0.060)	(-0.030, 0.110)
logo3	0.050	0.050	0.060	0.070
	(-0.146, 0.246)	(-0.166, 0.266)	(-0.156, 0.276)	(-0.146, 0.286)
	(-0.460, 0.570)	(-0.470, 0.570)	(-0.460, 0.580)	(-0.450, 0.590)
	(-0.010, 0.120)	(-0.020, 0.120)	(-0.010, 0.130)	(0.000, 0.140)
lognox	0.000	0.030	0.070	0.090
	(-0.216, 0.216)	(-0.166, 0.226)	(-0.126, 0.266)	(-0.106, 0.286)
	(-0.520, 0.520)	(-0.490, 0.550)	(-0.450, 0.590)	(-0.430, 0.610)
	(-0.070, 0.070)	(-0.040, 0.100)	(0.000, 0.140)	(0.020, 0.160)
logso2	0.210	0.230	0.240	0.260
	(-0.046, 0.466)	(-0.026, 0.486)	(-0.034, 0.514)	(-0.014, 0.534)
	(-0.310, 0.720)	(-0.290, 0.740)	(-0.270, 0.760)	(-0.260, 0.780)
	(0.140, 0.270)	(0.160, 0.290)	(0.180, 0.310)	(0.190, 0.330)

Table 3: Extension results with doughnut trimming and bias-aware confidence intervals

Notes: The four regressions are the same as in the previous section (in R with the GLM package). The first set of confidence intervals are based on the standard errors clustered in blocks of 35 days. Bias-aware confidence intervals are reported for $M = 1$ for the second set of confidence intervals and $M = 0.1$ for the third set of confidence intervals.

3 Extension 2 Optimal Bandwidth

As pointed out by Salas (2010) and Gallego et al. (2013) the choice of the time window (8 years) by Davis (2008) was a random choice. Davis (2008) did not justify this choice and as illustrated by Salas (2010) this choice has a profound impact on the results of the policy. The large bandwidth includes observations far from the cutoff that behave differently than observations close to the cutoff. This arguably leads to a wrongly estimated functional form and hence biased results at the cutoff. In this extension we take into account the optimal bandwidth for each pollutant as suggested by Imbens & Kalyanaraman (2012) and which was also implemented in the replication of Gallego et al. (2013).

The optimal bandwidth as proposed by Imbens & Kalyanaraman (2012) is calculated based on minimizing the asymptotic mean squared error of the RD estimator by using plug-in estimates for curvature, variance, and density, and applying regularization to stabilize the estimate. The logic behind this method is to optimize the trade-off between bias and variance when selecting the bandwidth. A larger bandwidth reduces variance but increases bias, while a smaller bandwidth reduces bias but increases variance. The derived formula provides the bandwidth that balances these two forces to minimize the overall mean squared error:

$$h_{\text{opt}} = C_K \left(\frac{\sigma_+^2(c) + \sigma_-^2(c)}{f(c) \left(m_+^{(2)}(c) - m_-^{(2)}(c) \right)^2} \right)^{1/5} N^{-1/5}$$

In this equation, C_K is a constant from the chosen kernel, $\sigma_+^2(c)$ is the variance after the cutoff, $\sigma_-^2(c)$ is the variance before the cutoff. Furthermore, $f(c)$ is the density of the running variable at the cutoff, $m_+^{(2)}(c)$ and $m_-^{(2)}(c)$ are the second derivatives (curvatures) of the regression functions on either side of the cutoff, and N is the sample size.

In the part below, we will implement the optimal bandwidth for all pollutants consistently reported with robust clustered confidence intervals, and bias aware confidence intervals for $M = 1$ and $M = 0.1$. We used the package `rdrobust` in R, which was developed by Calonico et al. (2014), for computing the optimal bandwidth according to the method of Imbens & Kalyanaraman (2012).

Outcome	Bandwidth	Bandwidth (Doughnut)
logco	202.600	214.570
logno2	78.620	43.520
logo3	108.110	101.700
lognox	90.980	80.260
logso2	92.020	90.160

Table 4: Optimal Bandwidth with and without Doughnut Trimming (1 week)

As shown in the table above, the optimal bandwidth when optimizing the bias-variance tradeoff with and without doughnut trimming is around 7 months for CO, around 2,5 months for NO₂ (a bit more than one month after doughnut trimming), around three months for O₃, a bit less than three months for NO_x, and around three months for SO₂. These values are much lower than the time window of eight years used by Davis (2008).

Outcome	Davis (GLM)	OB	OB (1 week)
logco	0.010	-0.368	-0.379
	(-0.166, 0.186)	(-0.424, -0.312)	(-0.435, -0.323)
	(-0.510, 0.520)	(-0.914, 0.178)	(-0.925, 0.167)
	(-0.060, 0.070)	(-0.469, -0.279)	(-0.464, -0.272)
logno2	-0.060	-0.089	-0.191
	(-0.334, 0.214)	(-0.181, 0.003)	(-0.319, -0.063)
	(-0.570, 0.460)	(-0.665, 0.486)	(-0.797, 0.416)
	(-0.120, 0.010)	(-0.216, 0.037)	(-0.347, -0.034)
logo3	0.050	-0.301	-0.332
	(-0.146, 0.246)	(-0.389, -0.213)	(-0.422, -0.242)
	(-0.460, 0.570)	(-0.874, 0.271)	(-0.908, 0.243)
	(-0.010, 0.120)	(-0.424, -0.179)	(-0.466, -0.215)
lognox	0.000	-0.629	-0.673
	(-0.216, 0.216)	(-0.739, -0.519)	(-0.787, -0.559)
	(-0.520, 0.520)	(-1.219, -0.039)	(-1.268, -0.077)
	(-0.070, 0.070)	(-0.786, -0.505)	(-0.815, -0.526)
logso2	0.210	-0.112	-0.123
	(-0.046, 0.466)	(-0.182, -0.042)	(-0.193, -0.053)
	(-0.310, 0.720)	(-0.669, 0.445)	(-0.681, 0.434)
	(0.140, 0.270)	(-0.231, -0.016)	(-0.225, -0.011)

Table 5: Extension results with Optimal Bandwidth and Bias-Aware Confidence Intervals

Notes: Davis (GLM) is the replication of Davis in R with the GLM package. OB is the replication of Davis under the optimal bandwidth. OB (1 week) is the replication of Davis with the doughnut trimming of one week under the optimal bandwidth. Each cell contains the coefficient, followed by three different confidence intervals (robust clustered SE, bias aware CI for M=1, bias aware CI for M=0.1)

The point estimates obtained using the optimal bandwidths without doughnut trimming are all significantly negative based on robust clustered standard errors and bias-aware confidence intervals with $M = 0.1$, except for NO₂. CO significantly declined by 36.8 percent, O₃ by 30.1 percent, NO_x by 62.9 percent, and SO₂ by 11.2 percent. Nonetheless, based on the conservative bias-aware confidence intervals with $M = 1$ only NO_x significantly declined.

The point estimates obtained using the optimal bandwidths with doughnut trimming are all significantly negative based on robust clustered standard errors and bias-aware confidence

intervals with $M = 0.1$, also for NO₂. CO significantly declined by 37.9 percent, NO₂ by 19.1 percent, O₃ by 33.2 percent, NO_X by 67.3 percent, and SO₂ by 12.3 percent. Nonetheless, based on the conservative bias-aware confidence intervals with $M = 1$ only NO_X significantly declined.

The point estimates obtained with and without doughnut trimming lie within each other's confidence intervals. Consequently, they cannot be considered statistically different from each other.

4 Extension 3 Semi-Parametric approach

A semi-parametric approach combines parametric assumptions to account for natural pollution covariates and non-parametric to capture the time-in-varying component. This approach is distinct from the previous two extensions as it relaxes some of the theoretical assumptions made about the data (Salas 2010). The residual series is regressed on time through a Kernel Local weighted regression using as weights kernel function Epanechnikov. The model to be estimated will be of the form

$$y_t = \beta_0 + B_2 x_t + \lambda(t) + \mu$$

where $\lambda(t)$ is a scalar function of time to be determined non-parametrically and x_t is the usual vector of covariates. The non-parametric component is estimated through a local average fit which may be less accurate than a true parameter but is more flexible and simple to use compared to other methods. There are also other semi-parametric methods addressed in the literature on Davis like a Lowess estimator that uses a tricubic kernel rather than an Epanechnikov kernel (Salas, 2010). The main value of the third extension is that it allows for an external check on the reliability of our results from the previous two extensions under less underlying assumptions of the data. As shown in the table below, despite using a different econometric methodology that relied more heavily on data, the results are still in line with the previous extensions which improves confidence that the estimated impact of the HNC policy is closer to the true impact.

The Kernel function Epanechnikov has the following structure. $\frac{3}{4}(1-z^2) \times (1|z < 1)$ where $(1|z < 1)$ is an indicator function that takes the value of 1 if the condition in the parantheses is satisfied (Salas, 2010). This function is reportedly the standard choice so as an arbitrary choice (which kernel method to choose), they were the option chosen as an external check on our results. While running the code in R, there were multiple errors in multi linearity which may be a cause of concern even though the code still executed and the results were as expected.

While the results in the table below on the Kernel regression results are discussed comprehensively in the next section, there does seem to be a difference in the result when donut trimming was applied except for Nitrogen Oxide which lends support that there was measure-

ment error at the discontinuity. Furthermore, the table also show how the assumption of the bounds at the discontinuity 'jump' can have inconclusive results by changing the absolute size of the confidence intervals.

Outcome	Davis (GLM)	OB (ND)	Kernel (ND)	OB (D)	Kernel (D)
logco	0.010	19.260	-0.434	17.930	-0.298
	(-0.166, 0.186)		(-0.532, -0.336)		(-0.394, -0.202)
	(-0.510, 0.520)		(-0.999, 0.131)		(-0.923, 0.237)
	(-0.060, 0.070)		(-0.567, -0.302)		(-0.427, -0.169)
logno2	-0.060	11.790	-0.501	12.570	-0.171
	(-0.334, 0.214)		(-0.727, -0.275)		(-0.325, -0.017)
	(-0.570, 0.460)		(-0.836, 0.379)		(-0.795, 0.416)
	(-0.120, 0.010)		(-0.749, -0.254)		(-0.353, 0.011)
logo3	0.050	26.560	-0.340	27.210	-0.286
	(-0.146, 0.246)		(-0.425, -0.255)		(-0.360, -0.212)
	(-0.460, 0.570)		(-0.838, 0.287)		(-0.832, 0.301)
	(-0.010, 0.120)		(-0.397, -0.155)		(-0.399, -0.174)
lognox	0.000	26.680	-0.597	25.390	-0.609
	(-0.216, 0.216)		(-0.695, -0.499)		(-0.717, -0.501)
	(-0.520, 0.520)		(-1.198, 0.000)		(-1.067, 0.141)
	(-0.070, 0.070)		(-0.729, -0.465)		(-0.750, -0.469)
logso2	0.210	28.970	0.045	26.010	0.007
	(-0.046, 0.466)		(-0.012, 0.102)		(-0.073, 0.087)
	(-0.310, 0.720)		(-0.594, 0.533)		(-0.588, 0.558)
	(0.140, 0.270)		(-0.054, 0.143)		(-0.110, 0.123)

Table 6: Kernel regression results with and without doughnut trimming, Davis (GLM) results, and corresponding optimal bandwidths

Notes: This table reports Davis (GLM) coefficients, optimal bandwidths (OB), and coefficients for kernel regressions with (1 week) (D) and without doughnut trimming (ND). The first set of confidence intervals is based on robust clustered standard errors. The second set corresponds to bias-aware confidence intervals with $M = 1$, and the third set corresponds to bias-aware confidence intervals with $M = 0.1$.

5 Comparison of the Extensions

Outcome	Davis (GLM)	Kernel (ND)	OB (ND)	D, OB, BA CI
logco	0.010	-0.434	-0.368	-0.379
	(-0.166, 0.186)	(-0.532, -0.336)	(-0.424, -0.312)	(-0.435, -0.323)
	(-0.510, 0.520)	(-0.999, 0.131)	(-0.914, 0.178)	(-0.925, 0.167)
	(-0.060, 0.070)	(-0.567, -0.302)	(-0.469, -0.279)	(-0.464, -0.272)
logno2	-0.060	-0.501	-0.089	-0.191
	(-0.334, 0.214)	(-0.727, -0.275)	(-0.181, 0.003)	(-0.319, -0.063)
	(-0.570, 0.460)	(-0.836, 0.379)	(-0.665, 0.486)	(-0.797, 0.416)
	(-0.120, 0.010)	(-0.749, -0.254)	(-0.216, 0.037)	(-0.347, -0.034)
logo3	0.050	-0.340	-0.301	-0.332
	(-0.146, 0.246)	(-0.425, -0.255)	(-0.389, -0.213)	(-0.422, -0.242)
	(-0.460, 0.570)	(-0.838, 0.287)	(-0.874, 0.271)	(-0.908, 0.243)
	(-0.010, 0.120)	(-0.397, -0.155)	(-0.424, -0.179)	(-0.466, -0.215)
lognox	0.000	-0.597	-0.629	-0.673
	(-0.216, 0.216)	(-0.695, -0.499)	(-0.739, -0.519)	(-0.787, -0.559)
	(-0.520, 0.520)	(-1.198, 0.000)	(-1.219, -0.039)	(-1.268, -0.077)
	(-0.070, 0.070)	(-0.729, -0.465)	(-0.786, -0.505)	(-0.815, -0.526)
logso2	0.210	0.045	-0.112	-0.123
	(-0.046, 0.466)	(-0.012, 0.102)	(-0.182, -0.042)	(-0.193, -0.053)
	(-0.310, 0.720)	(-0.594, 0.533)	(-0.669, 0.445)	(-0.681, 0.434)
	(0.140, 0.270)	(-0.054, 0.143)	(-0.231, -0.016)	(-0.225, -0.011)

Table 7: Comparison of Davis (GLM) estimates, Kernel (no doughnut) estimates, optimal bandwidth (no doughnut) estimates, and the estimates of our proposed corrections.

Notes: The first set of confidence intervals is based on robust clustered standard errors. The second set corresponds to bias-aware confidence intervals with $M = 1$, and the third set corresponds to bias-aware confidence intervals with $M = 0.1$.

The table above shows how different combinations of econometric approaches yield different impacts and different confidence intervals. The column Davis (GLM) displays the results of the initial analysis of Davis (2008) conducted in R with the package GLM. The kernel (ND) column shows the point estimates for the kernel regression without doughnut trimming we conducted in extension three. The OB (ND) column shows the point estimates of the regression with the optimal bandwidth according to the method of Imbens Kalyanaraman (2012) but without doughnut trimming. The final column reports the point estimates obtained using our proposed correction, which combines one-week doughnut trimming, bias-aware confidence intervals, and optimal bandwidth selection. The donut trimming was a correction to air quality measured by Mexican city sensors as an imprecise measure due to time lags, weather influences

and other factors. We applied the bias-aware confidence intervals to account for estimation bias and improve the accuracy of statistical inference. We also implemented the optimal bandwidth selection method proposed by Imbens and Kalyanaraman (2012) to better balance the bias-variance trade-off. This approach addressed a key limitation in Davis (2008), who did not account for bandwidth-induced bias in the regression discontinuity design.

The point estimates from the three extensions (Kernel, optimal bandwidth (OB), and our proposed correction) are largely consistent across pollutants. For all cases, the estimates are not significantly different from each other based on both robust clustered standard errors and bias-aware confidence intervals with $M=0.1$ and $M=1$.

Nonetheless, the point estimates from the three extensions are in almost all cases significantly different from the results found by Davis (2008). This suggests that the main issue in Davis's analysis lies in the choice of time window, a concern also raised by Salas (2010) and Gallego et al. (2013). Moreover, when considering only the robust clustered standard errors and bias-aware confidence intervals with $M=0.1$, all the point estimates become significantly negative except for SO_2 in the Kernel regression.

However, unlike Salas (2010) and Gallego et al. (2013), we find that using more conservative bias-aware confidence intervals with $M=1$ leads us to no longer reject the null hypothesis which is in line with the original results of Davis (2008). The only exception is NO_X for which the decline remains statistically significant.

Conclusion

This study critically evaluated the implications of measurement error in Regression Discontinuity (RD) designs. Considering there exists literature of influential papers with measurement issues and potentially invalid inferences, we hope this paper can be used as a warning to similar RDDs that ignore measurement error and a potential alternative solution. Davis work on HNC and air quality paper was cited by policy makers and replicated by researchers seeking to understand the environmental impact of a complicated policy with human reactions, changing weather and imprecise measurement of car emissions. In our replication, we showed that how our approach to address the measurement errors in air quality data resulted in different results from Davis' original paper.

In contrast to Salas (2010) and Gallego (2012), who also found a reduction in air pollution immediately following the HNC policy introduction, our paper had two distinct contributions. Firstly, we used our own unique econometric approach that was based on Davis and then integrated Bias-aware confidence intervals and donut trimming (from Dong and Kolesar's 2023 paper) alongside our own choice of optimal bandwidth correction. We also used a semi-parametric approach as an external validity check. Our results that were similar to Salas (2010) and Gallego (2012) suggest that Davis' paper once corrected for measurement error would have had similar results to other researchers working on the same problem and dataset.

The second contribution was that we suggest that results that showed a conclusive impact of the HNC on air pollution relied on certain underlying assumptions. Under very conservative bias adjustments, our results remained inconclusive, highlighting the sensitivity of RD estimates to the underlying assumptions about measurement error. However, by adopting more realistic data assumptions, our analysis uncovered a statistically significant reduction in air pollutants immediately following the implementation of the policy (HNC). This immediate improvement, however, was followed by a rebound in pollution levels over the ensuing months, with other researchers suggesting the cause was the purchase of older, less efficient vehicles. In summary, this replication effort showed the importance of correcting measurement error as well as a critical evaluation of assumptions that determine the confidence intervals.

Overall, our findings are relevant to future experiments with similar measurement error in RD analyses. Unlike other solutions, our empirical approach is more general and less taxing on data while maintaining conservative bias assumptions with the confidence intervals. We believe this approach leads to more valid inferences and a more critical understanding of the uncertainties in constructing non-standard confidence intervals.

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